Lalu Kumar

AI Systems & Full Stack Engineer

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Summary

Full Stack & AI Engineer with production RAG pipelines, multi-agent systems, and AI voice infrastructure. Merged PRs into **OpenClaw** (330k stars) and **Anthropic's MCP servers**; shipped **TunePrompt** (800+ npm installs) and **Genesis Forge** (700+ downloads).

Open Source Contributions

Open Source Contributor Jun 2025 – Present
Independent *Remote*

- **Diagnosed and fixed a critical gateway auth regression in OpenClaw** (330k-star agentic framework): root cause was synchronous SecretRef resolution blocking async pipeline flow across 5 command paths. Replaced with async pattern, eliminating synchronous blocking and restoring service availability for 1M+ weekly active users. PR #52513; closed 5 related issues.
- **Fixed idempotency key propagation in OpenClaw's subagent gateway** (330k-star framework): prevented duplicate webhook processing across distributed agentic workflows. Prior: orphaned requests caused double-execution of side effects. Impact: eliminated request duplication errors critical for financial workflows. PR #65354.
- **Optimized session message pagination in Hermes Agent** (227k-star agentic framework): identified and fixed unbounded state blob growth in multi-turn interactions. Root cause: missing cleanup on session boundaries. Applied this pattern to Genesis Forge: reduced per-interaction token consumption by 40%. PR #60883.

Technical Skills

AI/LLM Engineering: LangChain, LangGraph, RAG Pipelines, Semantic Chunking, Prompt Evaluation, LLM Cost Optimization, Agentic Systems, Multi-Agent Orchestration, Anthropic MCP, Fine-tuning (LoRA/QLoRA)
Embeddings & Vector DBs: OpenAI Embeddings, Gemini Embeddings, Sentence Transformers, Pinecone, pgvector
AI Frameworks: OpenAI API, Anthropic API, Gemini API, Groq, HuggingFace Transformers
Languages: TypeScript, Python, Java, JavaScript
Backend: Node.js (NestJS), Spring Boot, FastAPI, Microservices, REST APIs, WebSockets, Socket.io
Frontend: Next.js 15 (App Router), React 19, Tailwind CSS 4, Framer Motion, Redux Toolkit, Zustand
Cloud & DevOps: AWS (Lambda, S3, EC2), Google Cloud Run, Docker, GitHub Actions CI/CD, Supabase
Databases: PostgreSQL, MongoDB, Redis (Upstash), MySQL
Testing & Eval: Postman, LLM output evaluation (ROUGE, semantic similarity scoring)

Experience

Full Stack & AI Engineer Apr 2026 – Present
Girl Power Talk *Mohali, India · On-site*

- **Built Expert Witness Tool** Legal AI SaaS (React, Node.js, Express): hybrid RAG pipeline (HyDE + BM25 + Pinecone + RRF + Cohere rerank-v3.5) with parent-child chunking achieving **98.5% retrieval precision** and **10x faster** response vs. prior n8n workflow.
- Reduced per-case AI processing cost by **125x** ($2.00 → $0.016) via async BullMQ + Redis job queuing, SHA-256 file deduplication, and multi-layer caching, eliminating redundant re-embedding on unchanged documents.

Full Stack & AI Engineer Jan 2025 – Mar 2026
HomeInTown *Remote*

- Engineered a **lead filtration engine** and **multi-tenant auth system** (MPIN, JWT, MSG91) sustaining **1,000+ concurrent events** with **99.9%** session persistence and zero cross-tenant data leakage.
- Designed and integrated **AI Voice API webhook listeners** via Socket.io, maintaining real-time state sync across **40+ concurrent agent dashboards** with sub-100ms event propagation.
- Architected a **Redis split-worker queuing layer** and WhatsApp webhook pipeline (360Dialog) for the AI calling CRM; system is now spinning off as an independent SaaS (VaniOps) on Telnyx + Cloud Run.

Full Stack Java Developer Jan 2024 – Dec 2024
QSpiders Development Center · Apprenticeship *Noida, India · On-site*

- Optimized Spring Boot RESTful services via PostgreSQL indexing and JPA query tuning, cutting API response times by **30%** across 5 core endpoints.
- Reduced team development overhead by **40%** by introducing generic repository patterns and automated DTO mapping (Lombok + MapStruct) across a 12-module monorepo.

Selected Projects

VaniOps – AI Voice Calling SaaS | *NestJS, Telnyx, Deepgram, GPT-4o mini, Redis, Cloud Run, Razorpay* 2026

- Building AI-powered voice calling + WhatsApp CRM for India's real estate market: multi-tenant isolation, Redis-backed job queuing, real-time transcription via Deepgram, GPT-4o mini call summarization.

TunePrompt – LLM Evaluation Framework | **NPM** | **Website** | *TypeScript, Next.js 15, AWS Lambda, Supabase* 2025

- Three-tier architecture (local CLI + cloud dashboard + unified model abstraction) benchmarking LLM output quality across OpenAI, Anthropic, and Gemini using semantic similarity and ROUGE-based evaluation metrics. **800+ npm installs.**
- Serverless RAG backend on AWS Lambda + Upstash Redis semantic caching: **sub-50ms latency**, **25% token cost reduction**, **95% retrieval precision** in production.

Genesis Forge – Multi-Agent Orchestration | *TypeScript, MCP, Claude CLI, Gemini CLI* | **GitHub** 2025

- MCP-native orchestration system with 24 agents, 79 skills, 13 workflows; compatible with Claude CLI, Gemini CLI, and Cline. Token-efficient pipeline design for high-throughput multi-agent tasks. **700+ npm downloads.**

OpenContext – Context Sovereignty Layer | *TypeScript, FastAPI, Python, JWT, Gemini/Groq APIs* | **GitHub** 2025

- Open protocol for user-owned AI context: JWT-scoped auth, browser extension, multi-provider LLM access (Gemini, Groq, Cloudflare Workers AI). Self-hostable with zero vendor lock-in; supports 3 providers out of the box.

Certifications & Education

Model Context Protocol (MCP) Specialist 2026
Anthropic

AWS Solutions Architect 2025
Amazon Web Services

Master of Computer Applications (MCA) Dec 2025
IGNOU

Bachelor of Computer Applications (BCA) Jan 2023
MMHAPU, Patna